Docker-02

Task:1

1.Create a tomcat container on 8080 and deploy sample application in tomcat.

Step :1

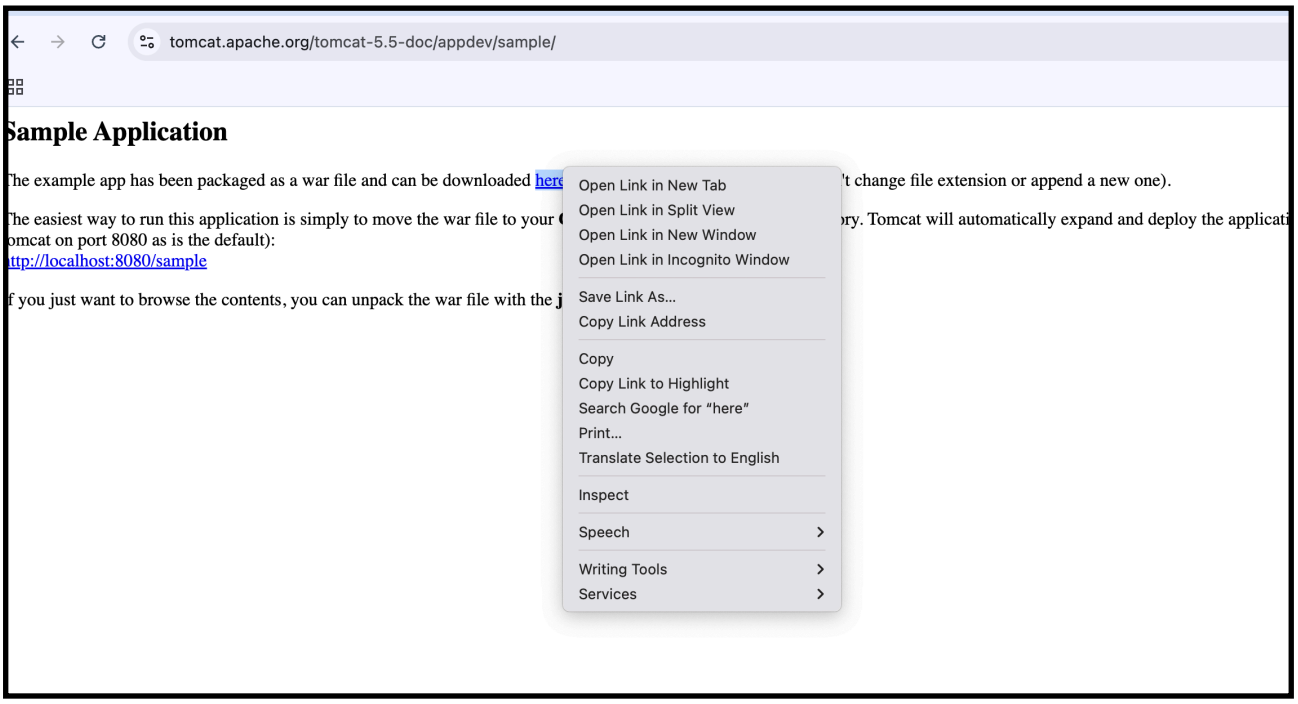
—> docker pull tomcat
* pull tomcat and run the image on 8080

—-> docker images
—-> docker run -itd -p 8080:8080 tomcat
—-> docker ps

Step:2

—> Deploy Sample Application (WAR file)

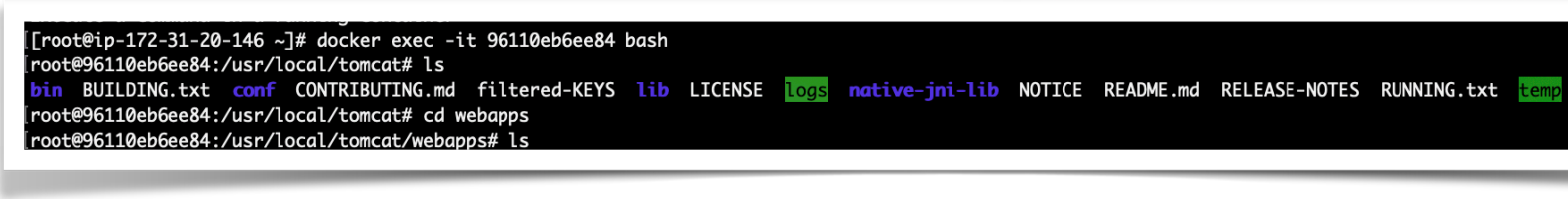
* go to official tomcat page of sample war



—> copy the link address

* now Use this command to go in to tomcat container

—> docker exec -it 96110eb6ee84 bash



—> cd webapps

Step:3

—> now use wget <link address which we copied >

-—> if you get the wget command not found then
—> apt update -y
—> apt install wget -y

```
root@96110eb6ee84:/usr/local/tomcat/webapps# wget https://tomcat.apache.org/tomcat-5.5-doc/appdev/sample/sample.war
bash: wget: command not found
root@96110eb6ee84:/usr/local/tomcat/webapps# apt update -y
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
```

```
root@96110eb6ee84:/usr/local/tomcat/webapps# apt install wget -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libpsl5t64 publicsuffix
The following NEW packages will be installed:
  libpsl5t64 publicsuffix wget
0 upgraded, 3 newly installed, 0 to remove and 6 not upgraded.
Need to get 520 kB of archives.
After this operation, 1,423 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libpsl5t64 amd64 0.21.2-1.1build1 [57.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 publicsuffix all 20231001.0357-0.1 [129 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 520 kB in 1s (660 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
```

—> now wget <url>

```
root@96110eb6ee84:/usr/local/tomcat/webapps# wget https://tomcat.apache.org/tomcat-5.5-doc/appdev/sample/sample.war
--2026-02-05 09:18:54--  https://tomcat.apache.org/tomcat-5.5-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: ‘sample.war’

sample.war                                          100%[=================================================================================>]

2026-02-05 09:18:54 (102 MB/s) - ‘sample.war’ saved [4606/4606]

root@96110eb6ee84:/usr/local/tomcat/webapps#
```

—> ls

```
root@96110eb6ee84:/usr/local/tomcat/webapps# ls
sample  sample.war
root@96110eb6ee84:/usr/local/tomcat/webapps# ll
total 8
drwxr-xr-x. 1 root root   38 Feb  5 09:18 ./
drwxr-xr-x. 1 root root   57 Jan 26 22:13 ../
drwxr-x---. 5 root root   86 Feb  5 09:18 sample/
-rw-r--r--. 1 root root 4606 Mar 31  2012 sample.war
root@96110eb6ee84:/usr/local/tomcat/webapps#
```

Step :4
—> now check your public ip:8080/sample

Task:2

2.Create volume and deploy tomcat container on port 8081.

1. Create a Docker Volume

`docker volume create tomcat-data`

This creates a named volume `tomcat-data`.

/var/lib/docker/volumes

```
[root@ip-172-31-20-146 ~]# docker volume create tomcat-data
tomcat-data
[[root@ip-172-31-20-146 ~]# cd /var/lib/docker/
[[root@ip-172-31-20-146 docker]# ls
buildkit  containers  engine-id  image  network  overlay2  plugins  runtimes  swarm  tmp  volumes
[[root@ip-172-31-20-146 docker]# cd volumes/
[[root@ip-172-31-20-146 volumes]# ls
259eeff72e272006b82922b8b653a46325a168f111d75d7ab524388c1177b26c  87752aa0f3145b9a1cebc003296651911a6012284d744a1ddd574bac423dca1e  backingFsBlockDev  metadata.db  tomcat-data
[[root@ip-172-31-20-146 volumes]# cd tomcat-data/
[[root@ip-172-31-20-146 tomcat-data]# ls
_data
[[root@ip-172-31-20-146 tomcat-data]# cd _data/
[[root@ip-172-31-20-146 _data]# ls
[[root@ip-172-31-20-146 _data]# pwd
/var/lib/docker/volumes/tomcat-data/_data
[root@ip-172-31-20-146 _data]#
```

2. Deploy Sample Application

Inside the /var/lib/docker/volumes/tomcat-data/_data

Deploy the sample war file in _data

—-> Wget <url>

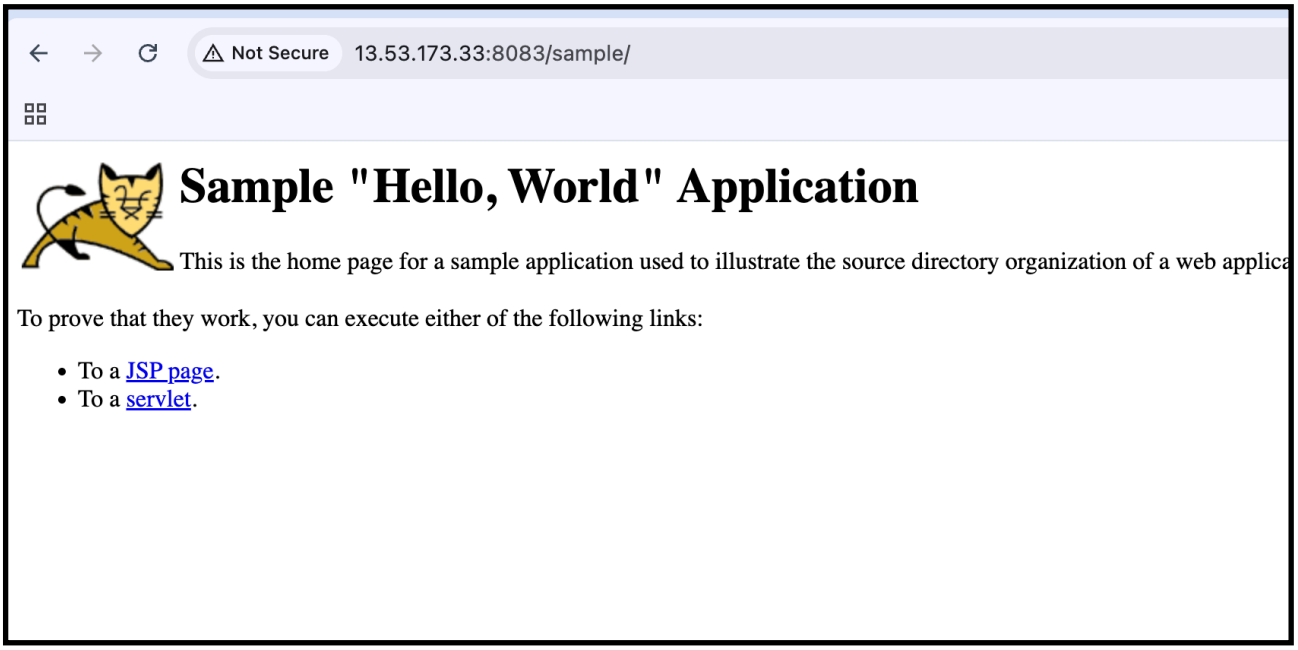
```
[[root@ip-172-31-20-146 _data]# pwd
/var/lib/docker/volumes/tomcat-data/_data
[[root@ip-172-31-20-146 _data]# wget https://tomcat.apache.org/tomcat-5.5-doc/appdev/sample/sample.war
--2026-02-05 09:34:12--  https://tomcat.apache.org/tomcat-5.5-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: ‘sample.war’

sample.war                                                  100%[=======================================================>]

2026-02-05 09:34:12 (102 MB/s) - ‘sample.war’ saved [4606/4606]

[root@ip-172-31-20-146 _data]# ls
sample.war
[root@ip-172-31-20-146 _data]# ll
total 8
-rw-r--r--. 1 root root 4606 Mar 31  2012 sample.war
[root@ip-172-31-20-146 _data]# ll
total 8
-rw-r--r--. 1 root root 4606 Mar 31  2012 sample.war
[root@ip-172-31-20-146 _data]#
```

3. Run Tomcat Container on Port 8083 with Volume

docker container run -itd -p 8083:8080 -v /var/lib/docker/volumes/tomcat-data/_data:/usr/local/tomcat/webapps tomcat:latest

```
[root@ip-172-31-20-146 _data]# docker container run -itd -p 8083:8080 -v /var/lib/docker/volumes/tomcat-data/_data:/usr/local/tomcat/webapps tomcat:lates
cf41d8af6cda0801022af1f75da9b92ecf6b253d71ef79e5d1a467322dbe1661
[root@ip-172-31-20-146 _data]# docker ps
CONTAINER ID   IMAGE          COMMAND             CREATED          STATUS          PORTS                                       NAMES
cf41d8af6cda   tomcat:latest  "catalina.sh run"   11 seconds ago   Up 11 seconds   0.0.0.0:8083->8080/tcp, :::8083->8080/tcp   hardcore_jones
96110eb6ee84   tomcat         "catalina.sh run"   25 minutes ago   Up 25 minutes   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp   keen_volhard
[root@ip-172-31-20-146 _data]# ll
total 8
drwxr-x---. 5 root root   86 Feb  5 09:36 sample
-rw-r--r--. 1 root root 4606 Mar 31  2012 sample.war
[root@ip-172-31-20-146 _data]#
```

—-> now check your public ip:8083/sample



Task:3

3.Limit the nginx container to 500 MB.

docker run -itd -p 81:80 --memory=500m nginx

—> What's happening here
-i -t -d → interactive + tty + detached (works, but -it isn't really needed for nginx)

-p 81:80 → host port 81 → container port 80

--memory=500m → limits RAM to 500 MB

nginx → image

```
[root@ip-172-31-20-146 ~]# docker run -itd -p 81:80 --memory=500m nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
0c8d55a45c0d: Pull complete
46bf3a120c8e: Pull complete
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
47cd406a84ef: Pull complete
bae5a1799a80: Pull complete
Digest: sha256:b17697e86d0c02378716277d09f45b946f8709aaa12c708e30fdd4736f536af1
Status: Downloaded newer image for nginx:latest
d283d67a9ebdab89831c243d3135a556fc93ffdc8c5a7f09ee0030c8f12b46f5
[root@ip-172-31-20-146 ~]# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                      NAMES
d283d67a9ebd   nginx          "/docker-entrypoint.…"   12 seconds ago   Up 11 seconds   0.0.0.0:81->80/tcp, :::81->80/tcp          serene_roentgen
cf41d8af6cda   tomcat:latest  "catalina.sh run"        8 minutes ago    Up 8 minutes    0.0.0.0:8083->8080/tcp, :::8083->8080/tcp  hardcore_jones
96110eb6ee84   tomcat         "catalina.sh run"        33 minutes ago   Up 33 minutes   0.0.0.0:8080->8080/tcp, :::8080->8080/tcp  keen_volhard
```

—> docker stats

```
[root@ip-172-31-20-146 ~]# docker stats

CONTAINER ID   NAME             CPU %     MEM USAGE / LIMIT     MEM %     NET I/O           BLOCK I/O     PIDS
d283d67a9ebd   serene_roentgen  0.00%     3.355MiB / 500MiB     0.67%     726B / 0B         0B / 32.8kB   3
cf41d8af6cda   hardcore_jones   0.10%     98.5MiB / 7.601GiB    1.27%     4.28kB / 5.62kB   0B / 221kB    35
96110eb6ee84   keen_volhard     0.05%     157.4MiB / 7.601GiB   2.02%     36.5MB / 127kB    0B / 64.5MB   35
```

Task :4 (task 5 - task 10)

4.Create a sample docker file using below instructions.

1. Base module as amazonlinux:latest. Task 5
2. Maintainer you name task 6
3. Install nginx task 7
4. COPY one index.html file to image. Task 8
5. Expose on port 80 task 9
6. Command to start the nginx container task 10

Step :1

Requirements :

1. Create a file named **Dockerfile** with the above contents.

   Vi docker file

2.

Create `index.html`

```
[[root@ip-172-31-20-146 ~]# vi index.html
[[root@ip-172-31-20-146 ~]# cat
 .bash_history  .bash_logout   .bash_profile  .bashrc
[[root@ip-172-31-20-146 ~]# cat index.html
<!DOCTYPE html>
<html>
<head>
  <title>Custom Nginx</title>
</head>
<body>
  <h1>Hello from Amazon Linux Nginx</h1>
</body>
</html>

[root@ip-172-31-20-146 ~]#
```

Vi Dockerfile

```
[root@ip-172-31-20-146 ~]# vi dockerfile
[root@ip-172-31-20-146 ~]# cat dockerfile
# 5. Base module as amazonlinux:latest
FROM amazonlinux:latest

# 6. Maintainer name
MAINTAINER Waseem Akram

# 7. Install nginx
RUN yum update -y && \
    yum install -y nginx && \
    yum clean all

# 8. Copy one index.html file to image
COPY index.html /usr/share/nginx/html/index.html

# 9. Expose on port 80
EXPOSE 80

# 10. Command to start the nginx container
CMD ["nginx", "-g", "daemon off;"]

[root@ip-172-31-20-146 ~]#
```

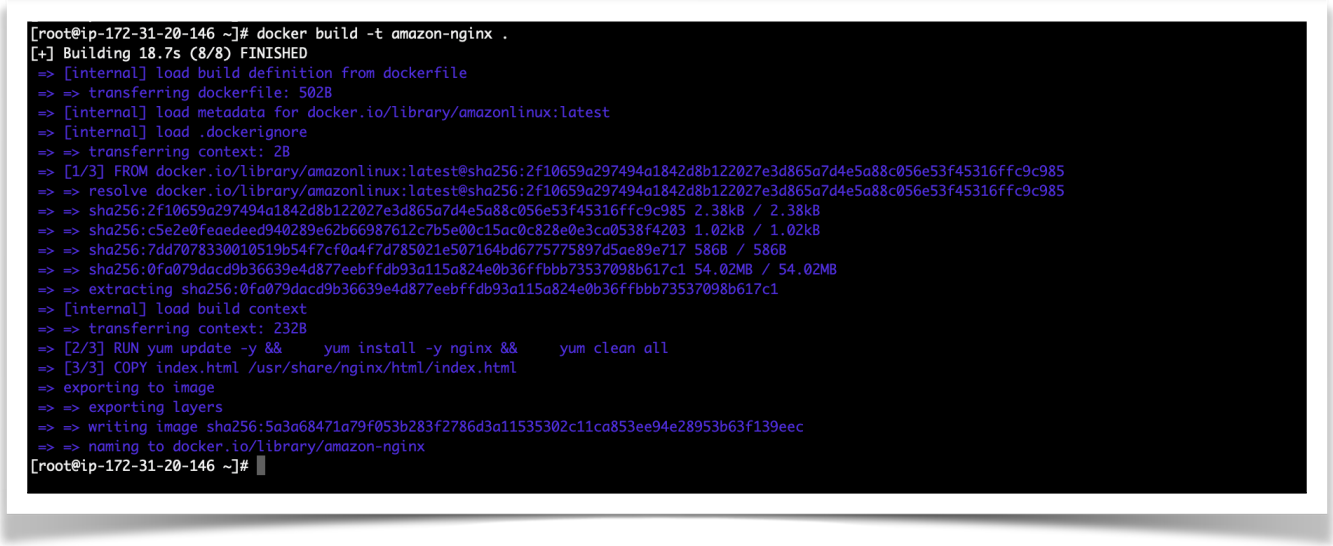
```
# 5. Base module as amazonlinux:latest
FROM amazonlinux:latest

# 6. Maintainer name
MAINTAINER Waseem Akram

# 7. Install nginx
RUN yum update -y && \
    yum install -y nginx && \
    yum clean all

# 8. Copy one index.html file to image
COPY index.html /usr/share/nginx/html/index.html

# 9. Expose on port 80
EXPOSE 80

# 10. Command to start the nginx container
CMD ["nginx", "-g", "daemon off;"]
~
~
~
~
~
```

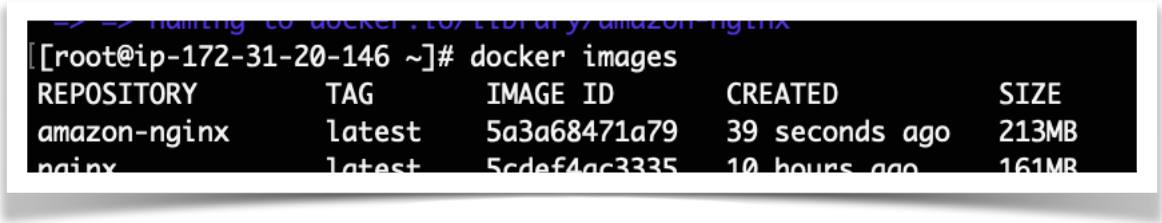
Step 2: Build the Docker image

```
docker build -t amazon-nginx .
```

Check:

```
docker images
```





Step 3: Run the nginx container

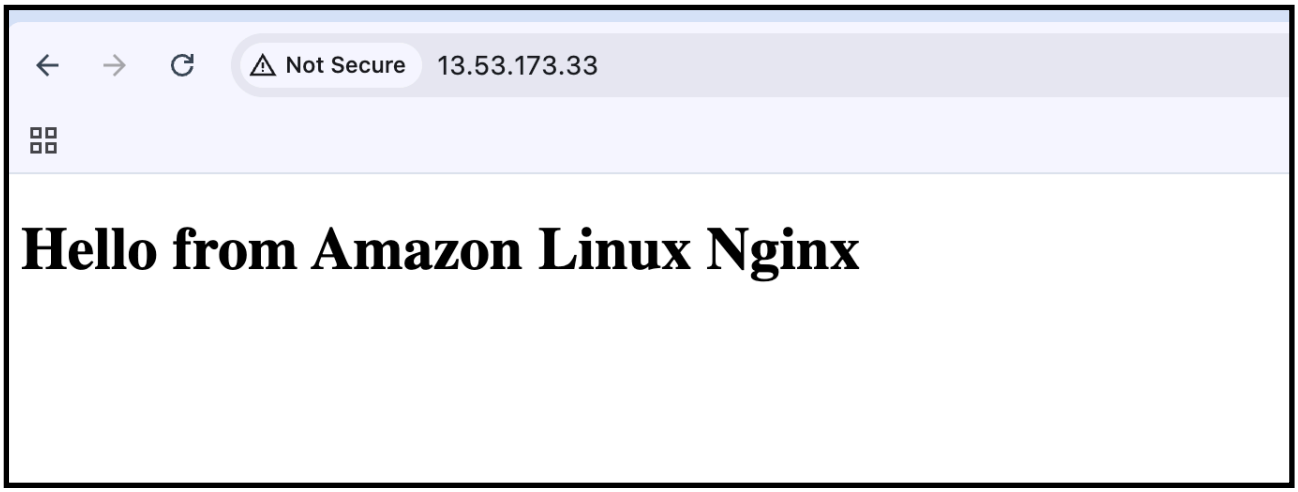
```
docker run -d -p 80:80 --name nginx-container amazon-nginx
```

Step 4: Verify

```
[root@ip-172-31-20-146 ~]# docker run -d -p 80:80 --name nginx-container amazon-nginx
cf284536f3f5caf7a90ca46bac5da83d17e3cee5af85cd38c1fb64a7ccc40d21
[root@ip-172-31-20-146 ~]# docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS          PORTS                                     NAMES
cf284536f3f5   amazon-nginx   "nginx -g 'daemon of…"   6 seconds ago    Up 5 seconds    0.0.0.0:80->80/tcp, :::80->80/tcp         nginx-container
d283d67e9ebd   nginx          "/docker-entrypoint.…"   20 minutes ago   Up 20 minutes   0.0.0.0:81->80/tcp, :::81->80/tcp         serene_roentgen
```

Browser:

`http://<SERVER-IP>`



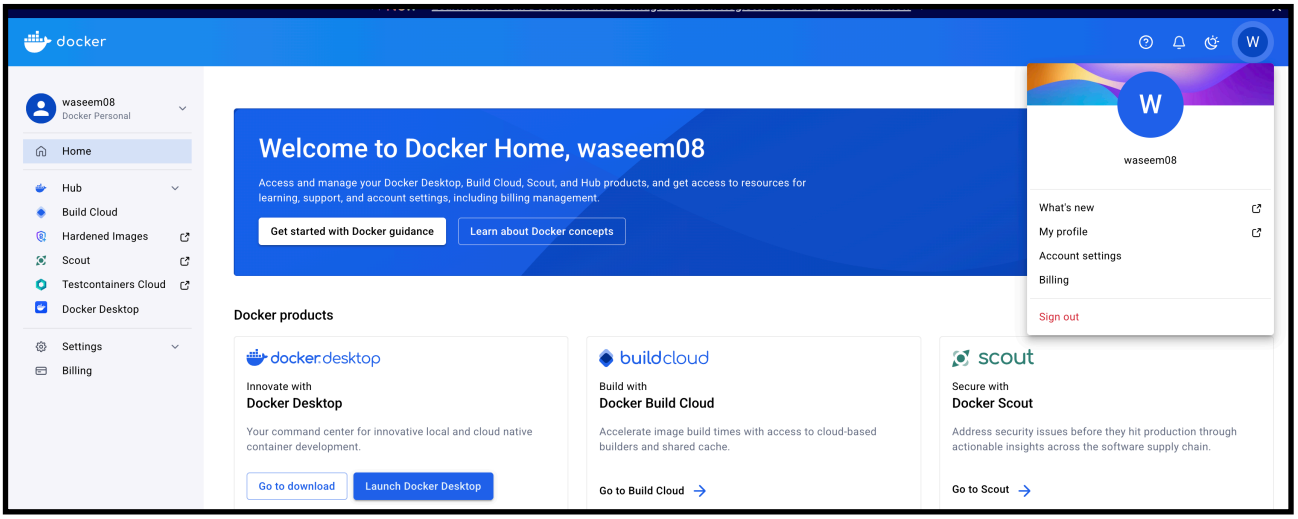
Task :11

11.Push image to Docker hub

Step 1: Login to Docker Hub

`docker login`

Enter your **Docker Hub username** and **password**.




```
[root@ip-172-31-20-146 ~]# docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub.
You can log in with your password or a Personal Access Token (PAT). Using a limited-sc

Username: waseem08
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
[root@ip-172-31-20-146 ~]#
```

Step 2: Tag the image

Your local image:

`amazon-nginx`
Tag it:

```
docker tag amazon-nginx waseem08/amazon-nginx:latest
```

```
Login Succeeded
[root@ip-172-31-20-146 ~]# docker tag amazon-nginx waseem08/amazon-nginx:latest
[root@ip-172-31-20-146 ~]# docker images
REPOSITORY              TAG       IMAGE ID       CREATED          SIZE
amazon-nginx            latest    5a3a68471a79   13 minutes ago   213MB
waseem08/amazon-nginx   latest    5a3a68471a79   13 minutes ago   213MB
```

Step 3: Push the image

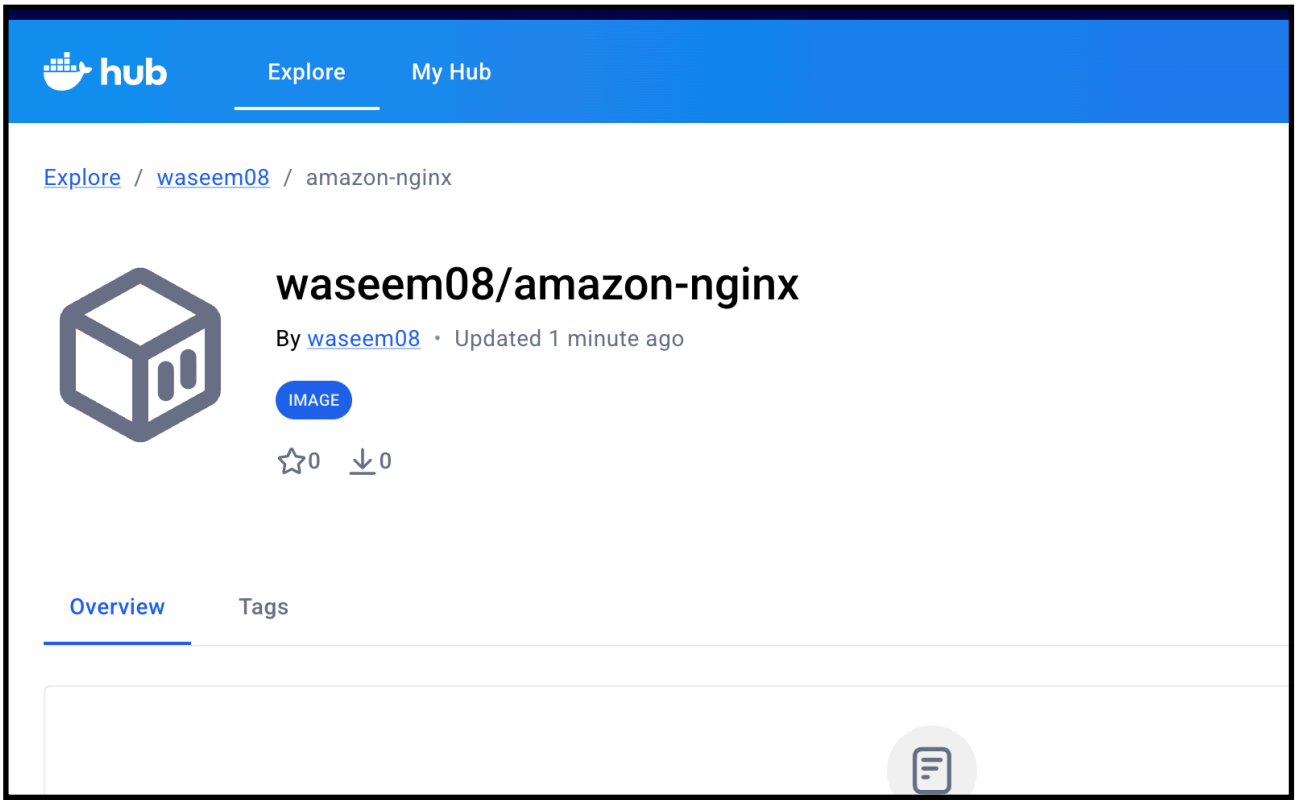
```
docker push waseem08/amazon-nginx:latest
```

```
[root@ip-172-31-20-146 ~]# docker push waseem08/amazon-nginx:latest
The push refers to repository [docker.io/waseem08/amazon-nginx]
721a997498da: Pushed
1ae283a870d9: Pushed
b535d73979ee: Mounted from library/amazonlinux
latest: digest: sha256:eba6433f9e1b8f2758fd2daa6790b6dffd8b034e1dc5efff5cefea3d348b69a0 size: 948
[root@ip-172-31-20-146 ~]#
```

Step 4: Verify

Go to 👉 https://hub.docker.com
Check **Repositories** → you'll see:

```
waseem08/amazon-nginx
```



Task:12

12.push image to AWS ECR

STEP 1: Configure AWS

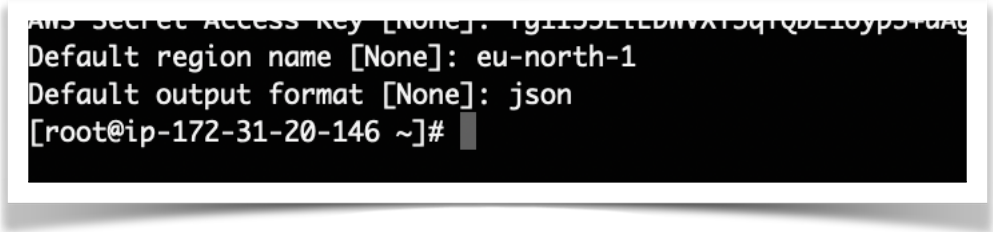
Run:

```
aws configure
```

You'll be asked **4 things**. Fill them like this:

```
AWS Access Key ID     : <your_access_key>
AWS Secret Access Key : <your_secret_key>
Default region name   : eu-north-1
Default output format : json
```

Use the **Access Key & Secret Key**

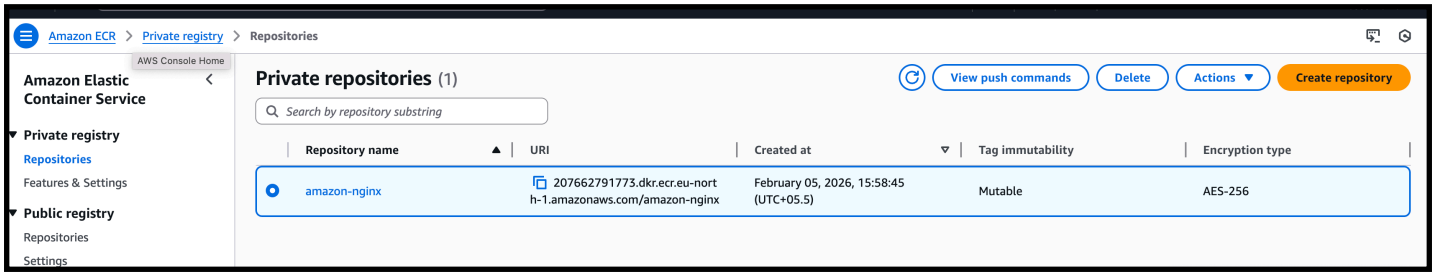


STEP 3: Create ECR repository

```
aws ecr create-repository --repository-name amazon-nginx --region eu-north-1
```

```
[root@ip-172-31-20-146 ~]# aws ecr create-repository --repository-name amazon-nginx --region eu-north-1
{
    "repository": {
        "repositoryArn": "arn:aws:ecr:eu-north-1:207662791773:repository/amazon-nginx",
        "registryId": "207662791773",
        "repositoryName": "amazon-nginx",
        "repositoryUri": "207662791773.dkr.ecr.eu-north-1.amazonaws.com/amazon-nginx",
        "createdAt": "2026-02-05T10:28:45.035000+00:00",
        "imageTagMutability": "MUTABLE",
        "imageScanningConfiguration": {
            "scanOnPush": false
        },
        "encryptionConfiguration": {
            "encryptionType": "AES256"
        }
    }
}
[root@ip-172-31-20-146 ~]#
```

—-> by this we have created the repository in our aws eu-north-1 region



STEP 4: Login Docker to ECR

```
aws ecr get-login-password --region eu-north-1 | docker login
--username AWS --password-stdin 207662791773.dkr.ecr.eu-
north-1.amazonaws.com
```

You should see:

```
Login Succeeded
```

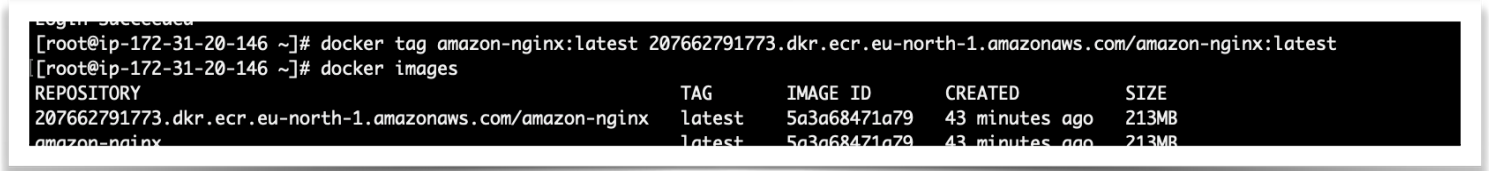
STEP 5: Tag your Docker image

Your local image:

```
amazon-nginx:latest
```

Command:

```
docker tag amazon-nginx:latest 207662791773.dkr.ecr.eu-
north-1.amazonaws.com/amazon-nginx:latest
```



STEP 6: Push image to AWS ECR

```
docker push 207662791773.dkr.ecr.eu-north-1.amazonaws.com/
amazon-nginx:latest
```

```
[root@ip-172-31-20-146 ~]# docker push 207662791773.dkr.ecr.eu-north-1.amazonaws.com/amazon-nginx:latest
The push refers to repository [207662791773.dkr.ecr.eu-north-1.amazonaws.com/amazon-nginx]
721a997498da: Pushed
1ae283a870d9: Pushed
b535d73979ee: Pushed
latest: digest: sha256:eba6433f9e1b8f2758fd2daa6790b6dffd8b034e1dc5efff5cefea3d348b69a0 size: 948
[root@ip-172-31-20-146 ~]#
```

Verify

AWS Console → **ECR** → **amazon-nginx** → **Images**

—-> now go and check your repository in aws ecr